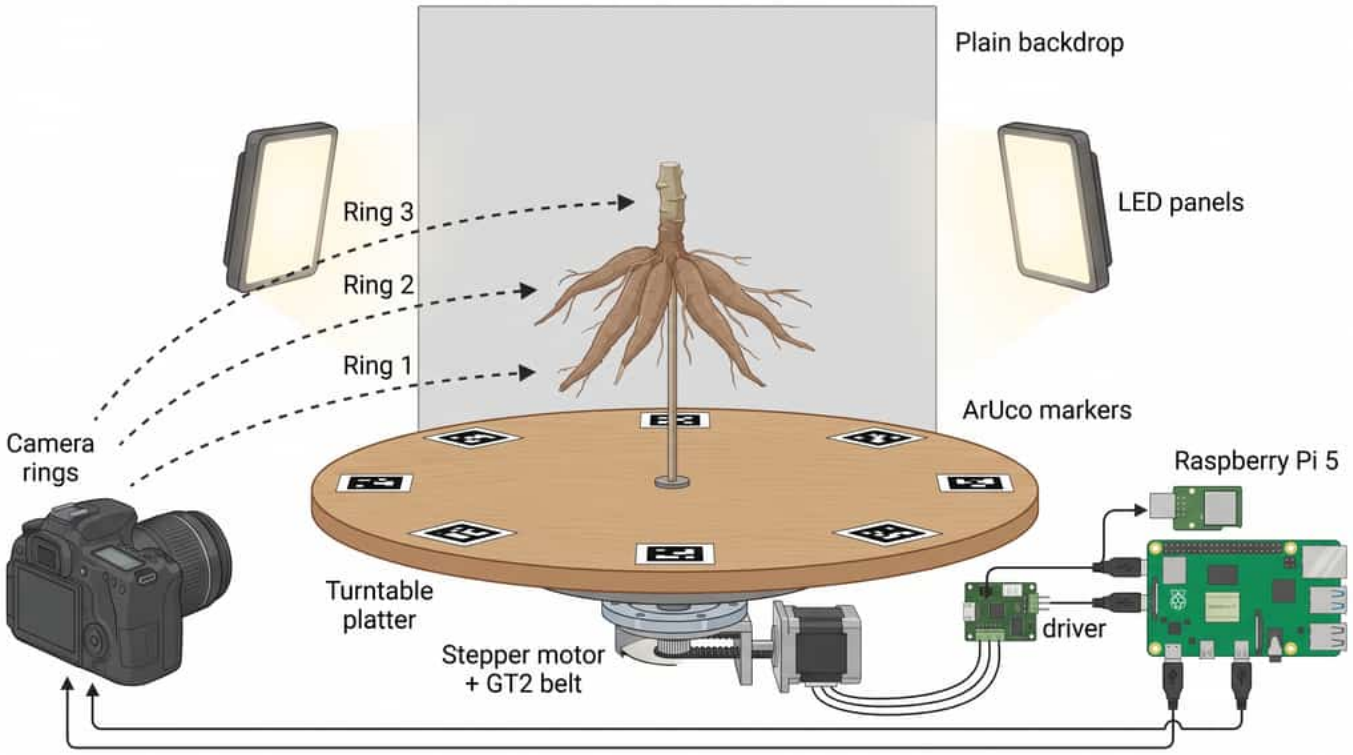
SOP-5 Motorised turntable

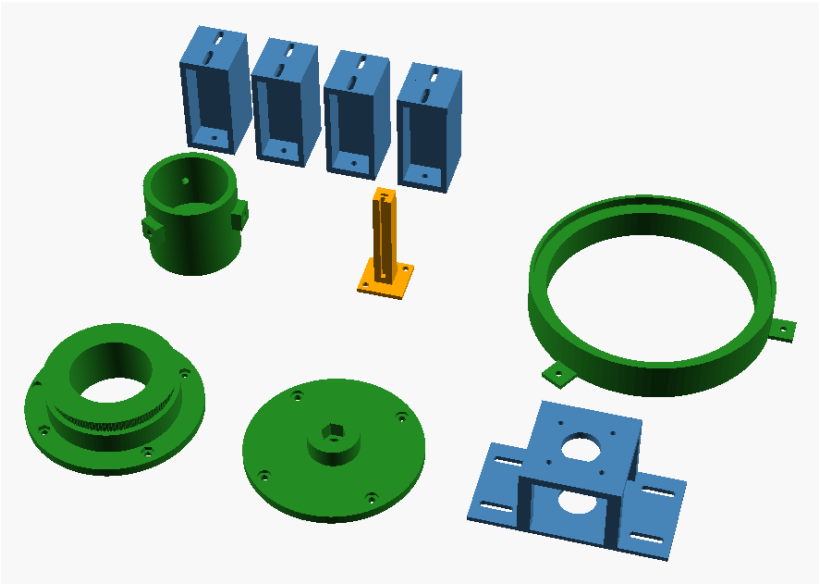
Build, print and run the PhenoForge turntable for root crowns and potted plants

SOP-5 · PhenoForge capture SOP · version 1.0 (October 2026)



The camera stays still and the sample turns in fixed steps. A Raspberry Pi 5 moves the platter and takes a photo at every stop, so every session is captured the same way. Design files are in **hardware/turntable**: the OpenSCAD source, STL files, Bambu Studio plates for the X1 Carbon, and the Pi script.

1 Print the parts



Printed parts: four risers, stem cup, hall-sensor post, pot ring, big pulley (120T), crown hub and motor cradle.

Plate (Bambu Studio file)	Parts	Settings (X1 Carbon, 0.4 mm)
plate1_drive_parts.3mf	big pulley, crown hub, motor cradle, stem cup, sensor post	PETG, 0.20 mm, 4 walls, 35 % gyroid, no supports, ≈250 g
plate2_risers_pot_ring.3mf	4 risers, pot ring (150 mm pot)	same, ≈265 g

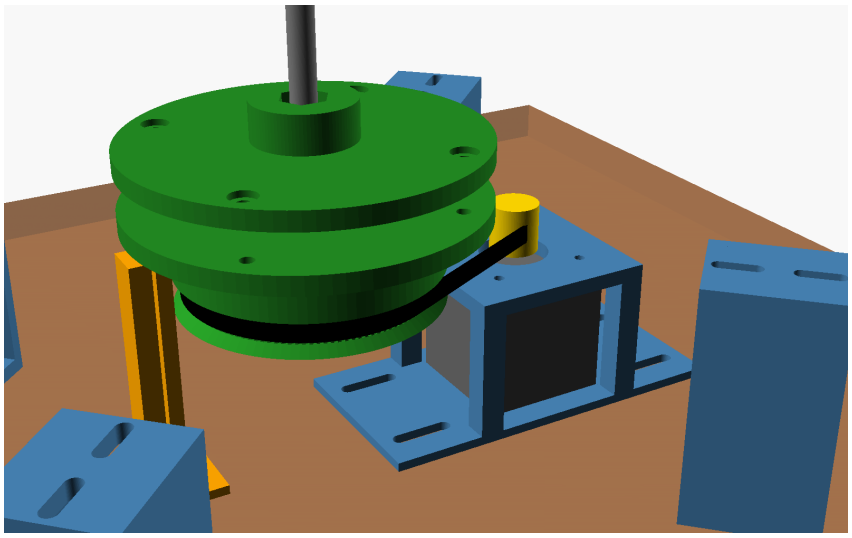
Open each 3MF in Bambu Studio, choose the *Bambu Lab X1 Carbon 0.4 nozzle* printer, PETG and the *0.20 mm Standard* process, set walls to 4 and infill to 35 % gyroid, then slice and print. To change a size (bearing, pot or stem diameter), edit the parameters at the top of **turntable.scad** and export again.

2 Buy the other parts

Shopping list (about AUD 120–160)

- 300 mm lazy-susan ball bearing, rated ≥ 100 kg, centre opening ≥ 120 mm
- NEMA 17 stepper (40 mm body) and a TMC2209 driver
- GT2 20T pulley (5 mm bore) and a GT2 closed belt, 300 mm, 6 mm wide
- 12 V 3 A power supply and a 100 µF capacitor
- Plywood base 500 × 500 × 18 mm; 450 mm platter, painted matte grey
- M8 threaded rod 250–300 mm, M4×40 bolts, M3×8 screws, M5 thumb screws
- Optional: A3144 hall sensor and a 6 × 3 mm magnet for homing
- Raspberry Pi 5 (your camera rig), DSLR on USB or the Arducam USB cameras

3 Assemble



Under the platter (platter and bearing hidden): the printed pulley hangs through the bearing opening, and the motor cradle slides outward to tension the belt.

1. **Base and bearing**
   Screw the 4 risers to the base on a 270 mm circle. Screw the bearing's lower ring onto the risers.

2. **Platter and pulley**
   Drill the platter centre: one 9 mm hole, and four 4.5 mm holes on a 94 mm circle. Bolt the crown hub on top and the big pulley underneath with 4 × M4×40, with the nuts in the pulley recesses. Screw the platter to the bearing's upper ring.

3. **Motor and belt**
   Fix the motor under the cradle plate and fit the 20T pulley and belt. Place the cradle about 73 mm from the centre, slide it out until the belt deflects about 3 mm under a light press, then screw down the slotted feet. Slide the 20T pulley on its shaft until the belt runs level.

4. **Wire the Pi**
   GPIO18 → STEP, GPIO23 → DIR, GPIO24 → EN, 3V3 and GND to the driver logic, 12 V to the driver motor supply (with the capacitor), and optionally the hall sensor to GPIO25. Set the driver current to about 70 % of the motor rating. Wiring details are in hardware/turntable/README.md.

5. **Mount the sample**
   Root crown: screw the M8 rod into the hub and clamp the stem stub in the stem cup with the 3 thumb screws. Potted plant: remove the rod and screw the pot ring to the platter.

6 **Markers**
Print 4–8 ArUco markers (Tools > Marker sheet, 40–50 mm) and glue them flat around the platter edge. They turn with the sample, which is what the reconstruction needs.

4 Capture a session

Setting	Root crown	Potted wheat
Stops per ring	36 (10°)	36 (10°)
Rings (camera heights)	3: below, level, 45° above	3: pot height, 30°, 60° above
Settle time after each move	1.5 s (heavy roots swing)	2–3 s (leaves and heads sway)
Speed and acceleration	15°/s, 15°/s²	10°/s, 10°/s²
Camera	DSLR f/8–f/11, ISO 100–400, manual focus and white balance	same; or both Arducams for two heights at once

1 **Test the drive**
`python3 pi/turntable_capture.py --jog 90`: the platter should turn a quarter turn smoothly. Use *--reverse* to change direction.

2 **Set up the camera**
Tripod 60–100 cm away, sample filling about 60 % of the frame, plain backdrop and two lights at 45°. Lock focus, exposure and white balance. Nothing else in view may move.

3 **Run**
`python3 pi/turntable_capture.py --out ~/scans/C07 --camera gphoto2 --shots 36 --rings low mid top --settle 1.5 --speed 15 --accel 15`. After each ring the script asks you to move the camera to the next height.

4 **Phone or other cameras**
Use *--camera opencv --cam 0 2* for the two Arducam USB cameras. Use *--camera none --settle 3* with a phone on an interval timer, then copy the photos into the session folder.

5 **Copy to the PC**
The session folder holds ring_index_angle.jpg photos and a turntable.json file. Copy the whole folder.

5 Reconstruct and measure in PhenoForge

1 **Import**
Tools > Import from a capture device, device *Turntable rig (session folder)*, **Photo folder…**. Blurred photos are rejected. Masks are then made from the room, which is identical in every photo, so COLMAP only uses the turning sample and the platter.

2 **Scale**
In the 3D analysis tab choose **ArUco markers** and type the measured marker size.

3 **Measure**
Root crowns: **Roots 3D** tab (SOP-4), stem direction +*z* or -*z*. Potted plants: **3D analysis**, plant mode *single*.

Troubleshooting

- Doubled leaves or roots: increase *--settle*, and lower *--speed* and *--accel*.
- COLMAP reconstructs the room or the backdrop: something static was mistaken for the sample. Check that nothing else moved, that the markers are on the platter (not the table), and that the backdrop is plain.
- The platter stalls or skips: raise the driver current a little, check the belt tension, and centre heavy samples.
- The motor hums but does not turn: the EN pin is not low or the coil pairs are swapped (swap A2 and B1).
- Wrong angle per stop: *--microsteps* does not match the driver's MS pins.